

TRIAPERFORMANCE GUIDE

The two thresholds and the “Zone 2” problem

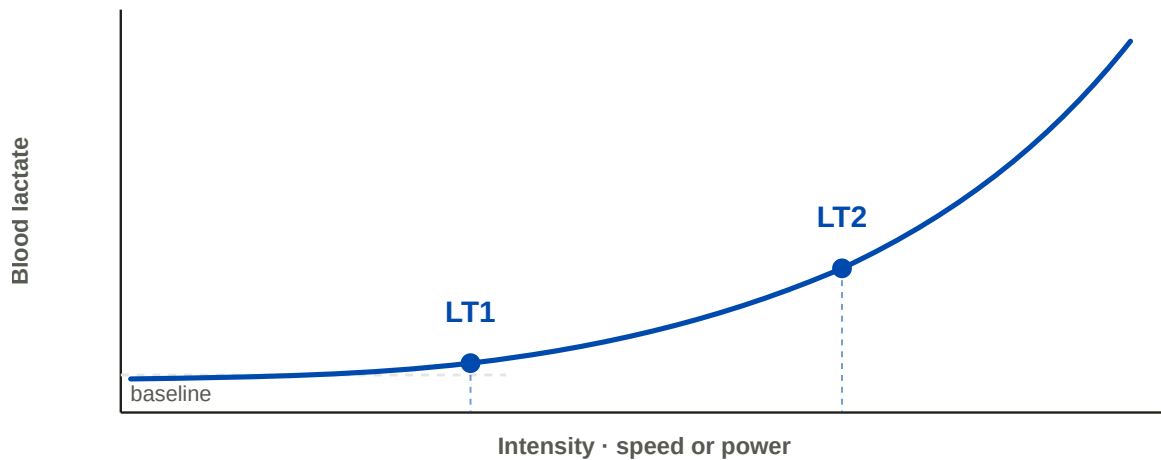
How lactate physiology, zone systems and the intensities you train relate to one another.



“Threshold” is two points, not one. And “Zone 2” does not mean the same thing in the 3-zone model as it does in the 5- or 7-zone one. This guide puts the physiology and the systems on a single axis — the percentage of your threshold — and shows where each piece fits.

There isn't one threshold. There are two.

As intensity rises — measured in heart rate, power or speed — blood lactate rises with it. That rise is not linear: it has **two inflection points**, and those two points are what we call thresholds.



The lactate curve and its two thresholds.

LT1 · Aerobic threshold

The first inflection point. Below LT1, lactate stays close to your baseline: production and clearance run at the same rate and nothing accumulates. This is the effort you can hold for hours.

Between LT1 and LT2

Lactate is now clearly rising: clearance no longer keeps up with production, so it starts to accumulate. But it does so slowly, and the effort holds for a good while. This band is where marathon pace and long-course race pace live.

LT2 · Anaerobic threshold

The second inflection point. Above LT2 the accumulation turns exponential: lactate climbs fast, fatigue arrives early, and the time you can hold the effort is measured in minutes.

Training moves the curve

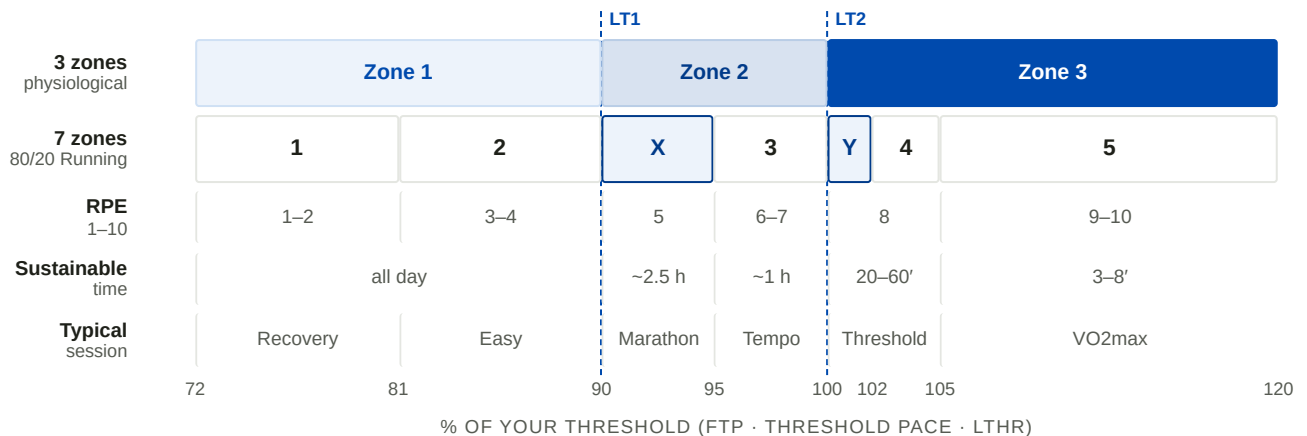
A trained athlete does not have a different curve — they have **the same curve shifted to the right**. They hold more speed or more watts before reaching each threshold. A large part of endurance training is exactly that.

From the lab to the field

Measuring LT1 and LT2 exactly requires a lab test. Since that is impractical, we use field protocols that estimate the **anaerobic threshold (LT2)** — your FTP, your threshold pace, your LTHR — and calculate the zones from there. When we say *threshold* on its own we mean LT2, and that number is **100%**.

Many systems, one physiology

There are many systems for setting training zones and prescribing intensity, and that is what confuses a lot of my athletes. The physiology is one thing; the zone systems are laid over it.



The systems on a single axis: the percentage of your threshold.

Where the two thresholds fall

LT2 sits at 100%: it is the number you estimate in the test, and everything else is calculated from it. LT1 falls around **90%**, with a real range of 85% to 95% depending on the athlete, their history and their sport — which is why it is drawn as a band rather than a line.

Three zones describe, seven prescribe

The 3-zone model cuts where the physiology changes regime, and it is what you use to talk about the *distribution* of a season. Seven zones are what you use to write Tuesday's session. One is the map, the other is the route.

Where Zone X and Zone Y come from

From the seven-zone system in *80/20 Running*, by Matt Fitzgerald. It adds two transition bands to the usual five: **X**, in the upper half of Zone 2, and **Y**, right around threshold. Separating them is what makes it possible to prescribe either one — or leave it out — on purpose. Page 7.

RPE and sustainable times are orders of magnitude for a trained athlete: they vary with the sport, the heat, your sleep and the week of the block. The zone bands are the same ones the calculator uses.

“Zone 2” doesn’t mean the same thing in every system

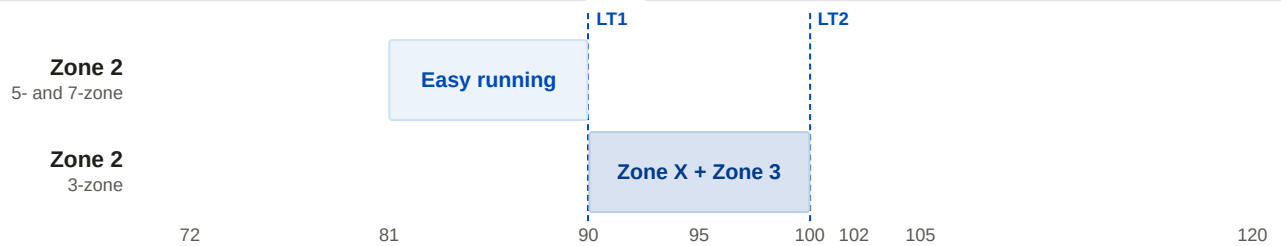
It is the overlap that generates the most questions. Both uses are correct inside their own model, and it is almost never stated which one is being used.

In the 3-zone model

Zone 2 is the whole band **between LT1 and LT2**: moderate intensity. Marathon pace, half-Ironman pace and tempo work all live there.

In the 5- and 7-zone models

Zone 2 sits entirely **below LT1**: it is the easy run, the one you hold for hours, where 80% of your volume lives.



The same label, two different bands. One begins exactly where the other ends.

The translation

Zone 2 in the 3-zone model is the same thing as **Zone X plus Zone 3** in the seven-zone one. Zone 2 in the seven-zone model sits below both, and takes no part in that conversation.

How to tell which one you are being given

If the system has three zones, its Zone 2 is the band between thresholds. If it has five or seven — your watch, your plan, the calculator — its Zone 2 is the easy run.

Find your threshold

Everything above needs a real number. These are the protocols we use, with the exact arithmetic for each one. None of them requires a lab.

SPORT	PROTOCOL	HOW IT IS CALCULATED
Running	30' at maximum sustainable effort	Threshold pace = the 30' average. LTHR = average HR of the last 20'.
Cycling · power meter	1 × 20' on a long climb	FTP = 95% of average power.
Cycling · short climb	2 × 8'	FTP = 90% of the average of both blocks.
Cycling · indoor	Ramp test	Simplest place to start. Overestimates slightly.
Cycling · no power meter	30' at maximum sustainable effort	LTHR = average HR of the last 20'.
Swimming	400 m + 200 m	CSS (critical swim speed) via calculator.

The rules that decide whether the test is worth anything

Early, cool, rested, fed and hydrated. Flat, uninterrupted route — one traffic light invalidates the test. Chest strap required. And “**maximum sustainable**” is not “**maximum**”: the effort has to reach the end intact.

You have your number. What about the zones?

The calculator builds them free, for all three sports, with all seven zones and all three metrics:
triaperformance.com/en/training-zones-calculator

RPE and sustainable time

RPE — perceived effort

RPE (a 1 to 10 scale) **integrates what no device measures**: how much you slept, how much stress you are carrying, what you ate, how hot it is. It tracks your relative intensity closely, with the caveat that the relationship shifts with duration, heat and the state of your glycogen stores.

In practice. A threshold session should feel around **RPE 8**. If one day your usual watts feel like a 10, the 10 is the valid data point: bring the intensity down and hold the physiological target, not the prescribed number. It is the correction our athletes learn to make on their own, in the moment.

In tests we prescribe effort rather than pace, so the result is today's and not the previous test's.

TTE — time to exhaustion

TTE is how long you can hold your threshold *today*. Two athletes with the same FTP are not the same athlete: one holds it for 25 minutes, the other for 55. In trained athletes it usually falls between **30 and 70 minutes**.

This is what explains why Zone 4 exists. The goal of threshold work is not only to raise the number — it is to **stretch how long you can hold it**. That is why a threshold session is prescribed as accumulated time in zone, and why that time grows gradually across the block.

What each RPE feels like

The cheapest way to calibrate RPE is the talk test. It needs no battery and it works in all three sports.

RPE	WHAT IT FEELS LIKE	ZONE
1–2	You hold a whole conversation without noticing.	1
3–4	You speak in long sentences. You could go for hours.	2
5	You still talk, but you pick your moments.	X
6–7	Short sentences. Comfortably uncomfortable.	3
8	Three or four words. Sustainable, but you are counting the minutes.	4
9–10	Not a word.	5

How we use it

The physiology in the previous pages holds for any athlete. How the volume gets distributed across those zones is a training decision.

The distribution

Around **80% of the volume easy and 20% hard**. The exact split we write is **80/15/5** or **70/25/5** depending on what the race demands, and it turns pyramidal as the race approaches: the work shifts toward race pace and Zone 5 is held back.

The middle band is prescribed, not improvised

There is high-value training between LT1 and LT2: it is marathon pace, half-Ironman and Ironman pace, and it is where tempo work lives. Dosed well, it is one of the key sessions of the week. What costs you is arriving there without having planned it — you head out for an easy run, you feel good, the pace settles by itself, and you finish the week with the fatigue of a quality session that was never in the plan.

Zone X — marathon pace

The upper half of Zone 2. In a marathon or long-course block it is central content and it is trained on purpose. Outside those blocks we leave it out: it accumulates considerably more fatigue than low Zone 2 and returns a similar aerobic stimulus.

Zone Y — the band around threshold

The band surrounding threshold, between 100 and 102%. It is trained when it is exactly your race pace. To chase threshold adaptations we prescribe from 102% upward: threshold moves with fitness, rest and heat, and training right on the line leaves the session short on the days that number sits higher.

And now, which session grows each zone

This guide is the map; the next one is the route. “**The sessions that grow each zone**” covers, for all three sports, which session builds each of the seven and what not to do in each. Request it from the calculator.

Who we are

Triperformance is **Iván Koch’s** coaching practice: triathlon and running training built on data, not on templates. One service, the same methodology for everyone — the one you have just read — and athletes across the Americas and Europe, from a first 10k to Ironman.

46

REVIEWS · ALL 5 STARS

328

PUBLISHED PLANS

3

LANGUAGES

“He provides insight into my Training Peaks data, helping me understand what the numbers mean.”

Sylmarie Arizmendi · Puerto Rico

“Always on top of my training and available to make any adjustments I needed.”

Giles Carmichael · United Kingdom

THE NEXT STEP

The map is there. The route is what's missing.

Knowing where your thresholds are is the starting point. What changes the outcome is a well-distributed week, held for months, that adjusts when life gets in the way. Three ways to make that happen.

RECOMMENDED

US\$ 39.99/month

All-Access Membership

Every training plan in the catalogue, with these zones and these sessions already sequenced week by week. Switch plans as often as you like.

- All 328 plans: running, cycling, triathlon, HYROX
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